Side-Jet Project

Side-Jet Interactions over a Hypersonic Cone Flow

Mehmet Furkan Tunç
06 December 2025

1. Introduction

Karpuzcu and Levin [1] studied side-jet interactions over a hypersonic cone flow using DSMC method.

In this report,

- How they conducted their studies,
- What are the results,
- How we can conduct our study from their results

are presented.

[1] Karpuzcu, I. T., & Levin, D. A. (2023). Study of side-jet interactions over a hypersonic cone flow using kinetic methods. AIAA Journal, 61(11). https://doi.org/10.2514/1.J063020

2. Study Inputs

There are two pillars to this study as given below:

Side-jet Study

Time-averaged Flowfield

- Mach contours, shock structures
- Penetration Heights, Separation Lengths
- Surface Pressure Distributions
- Total Drag Force, Normal Force, Heat Transfer

Instantaneous Flowfield

- Unsteadiness Effects
- Strouhal Number Comparisons

2. Study Inputs

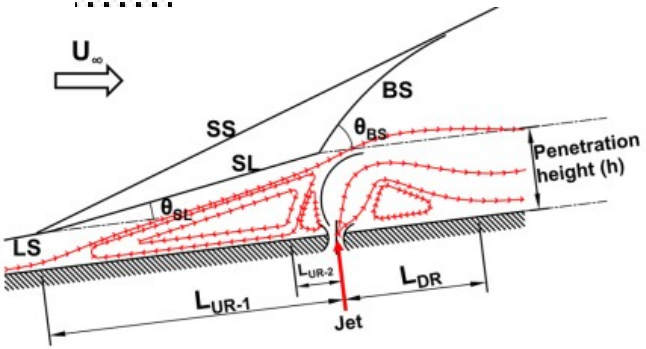


Fig. 4 Sketch of the interaction region showing shear-layer angle θ_{SL}, bow shock angle θ_{BS}, penetration height, and upstream (L_{UR-1}, L_{UR-2}) and downstream L_{DR} separation lengths.

Table 1 Freestream and jet flow parameters for the M=6 and M=12 case simulations

	M6		M12	
Parameter	Freestream	Jet	Freestream	Jet
Number density, m^{-3}	1.00×10^{22}	1.65×10^{23}	0.50×10^{22}	8.25×10^{23}
Mach number	6.0	1.0	12.0	1.0
Temperature, K	50.0	250.0	50.0	250.0
Pressure, Pa	6.9	571.2	3.45	2847.4
Knudsen number[a]	0.07	0.0067	0.15	0.0013

[a]Based on the reference length of the jet diameter of $d = 1$ mm.

Numerical Setup

Cone Diameter : 0.1 m
Cone Lenght : 0.4 m
Jet Diameter : 0.001 m
Freestream Re : 100000 m^{-1}
Normalizations are done with Cone Lenght
Jet's Center x/L : 0.5
Jet's Center y/L : 0.0625
Timestep : $5 * 10^{-9}$ s
Sampling Time: $5 * 10^{-4}$ s
Flow Establish Time : $2.5 * 10^{-3}$ s
Final Time : $5 * 10^{-3}$ s
Total Cost : 460000 CPU*hours

Physical Setup

Molecular Model : VHS
N_2 Molecule Dia. : $4.467 * 10^{-10}$ m
Viscosity Index : 0.26
Isothermal T_{wall} : 300 K
No Reactions
Axisymmetric Case
Both jet and freestream is N_2

3. Steady Study Outputs

There are two upstream separations and one downstream separation. Macroscopic scale has Mach contours and numerical Schlieren contours as given below:

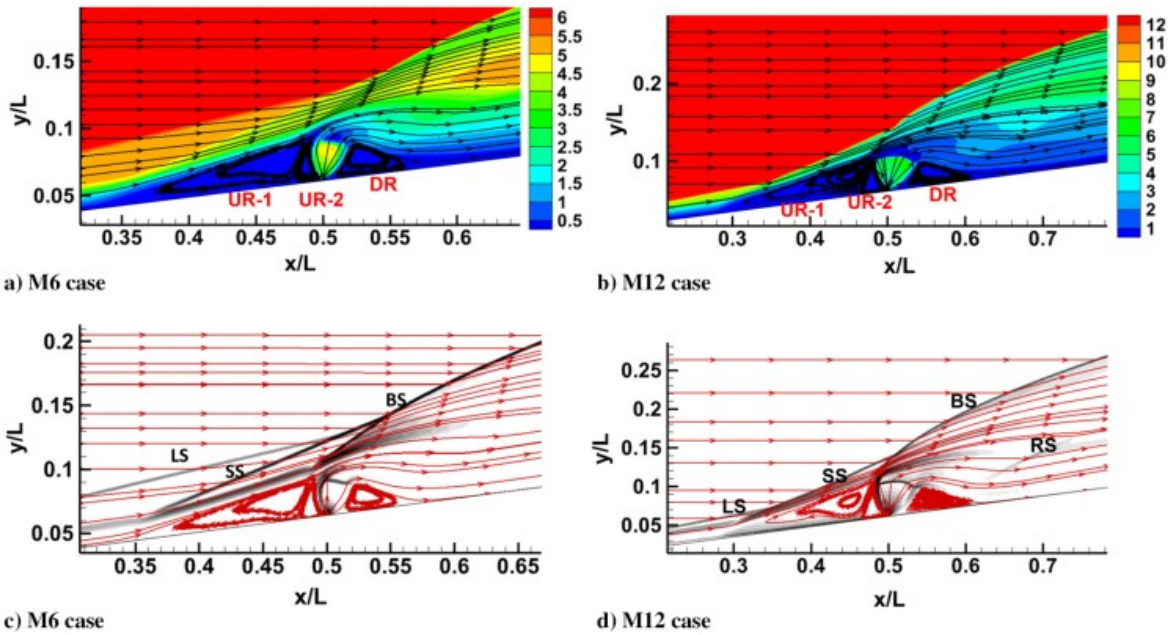


Fig. 2 Contours of Mach number and numerical schlieren for M6 and M12 cases.

3. Steady Study Outputs

Penetration heights, bow shock angles and amplification factors are necessary flow properties for accurately representing physics:

$$\widetilde{m} = (\dot{m}/4\rho_\infty U_\infty A_b)$$

Scaled Mass Flow Rate

$$h/r_b = 2\left[\frac{\gamma_j \widetilde{m}}{U_\infty}\right]^{1/2}\left[\frac{2R_j T_{0j}}{\gamma_j(\gamma_j - 1)}\right]^{1/4}$$

Theoretical Penetration Height

$$K_F = \frac{F_{\text{Interaction}} + F_{\text{Thrust}}}{F_{\text{VacuumThrust}}}$$

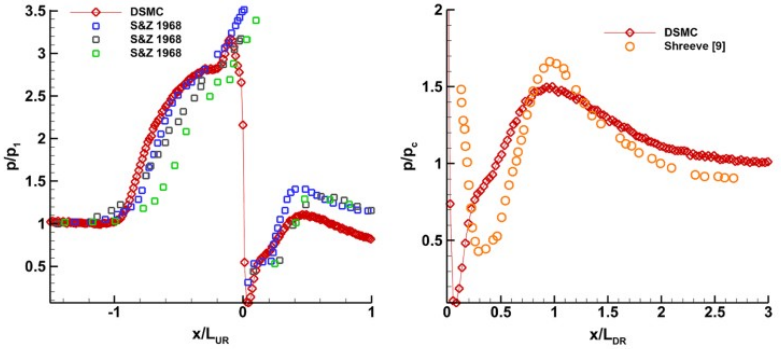
Amplification Factor

Table 3 Comparison of the parameters from DSMC with the literature

Parameter	M6 DSMC	M6 Literature [2,5,10]	M12 DSMC	M12 Literature [2,5,10]
Penetration height (h/r_b) [10]	0.32	0.331	0.5	0.523
Shear layer θ_{SI} and bow shock angles θ_{BS} [5]	17 and 33	17 and 36	13 and 31	13 and 33
Amplification factor [2]	5.9	2.5–5.2	2.89	2.5–5.2

Pressure distribution on the surface with different jet strengths*(60,100,200) are given on right side.

*Jet Strength: It is defined as the ratio of the jet total pressure to the static pressure of the incoming flow just upstream of separation



a) Comparison with Spaid and Zukoski [41] b) Comparison with Shreeve et al. [9]

Fig. 6 Comparison of surface pressures for M=6 case with the experimental results available in the literature [9,41]: DSMC simulations of this work (diamonds), and Shreeve et al. [9] (circles). Also, Spaid and Zukoski [41] is denoted as "S&Z 1968" in the legend.

3. Steady Study Outputs

At last, they compared results of integrated values of total drag force, normal force, and heat transfer:

Table 4 The ratio of surface forces with and without a side jet present. The quantities given here are the integral quantities over the cone surface

Force	Formula	M=6	M=12
Drag force ratio	F_{Djet}/F_{Dnojet}	0.76	0.68
Normal force ratio	F_{Njet}/F_{Nnojet}	2.30	1.74

Table 5 Heat transfer rates with and without a side jet present[a]

Jet presence	Property	M=6	M=12
Without jet	Total heat transfer rate W	12.6	249.2
With jet	Total heat transfer rate W	−5.68	59.3
With jet	Heat transfer rate due to jet gas W	−8.8	4.9
With jet	Heat transfer rate due to freestream gas W	3.12	54.4

[a]The quantities given here are the integral quantities over the cone surface.

4. Unsteadiness Study Outputs

They checked out flowfields at different instants and compared result differences to steady results:

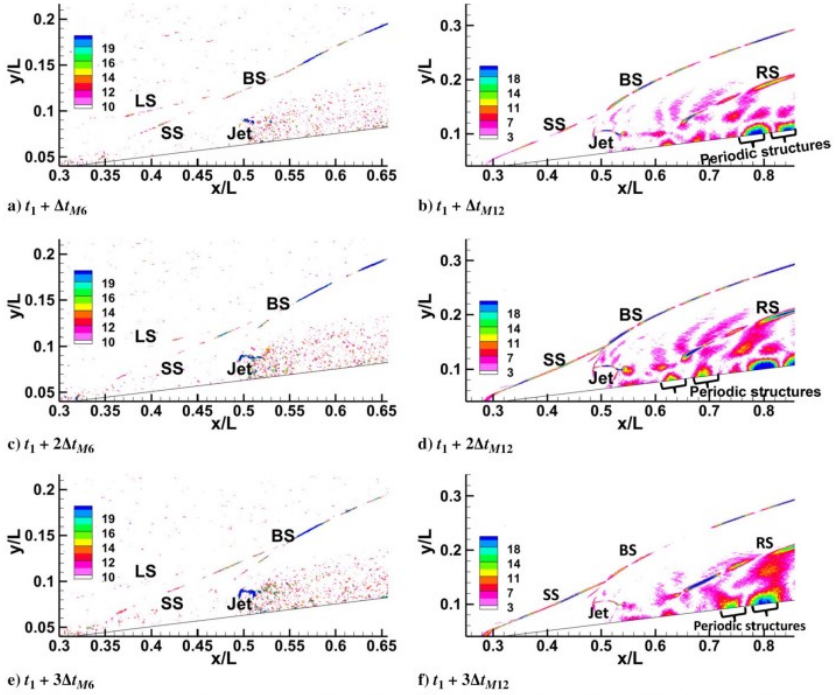


Fig. 8 Percent difference in pressure in instantaneous flowfields with indicated shock structures: M6 case (left column), and M12 case (right column).

They plotted flow velocity to flow oscillations using Ma and St numbers:

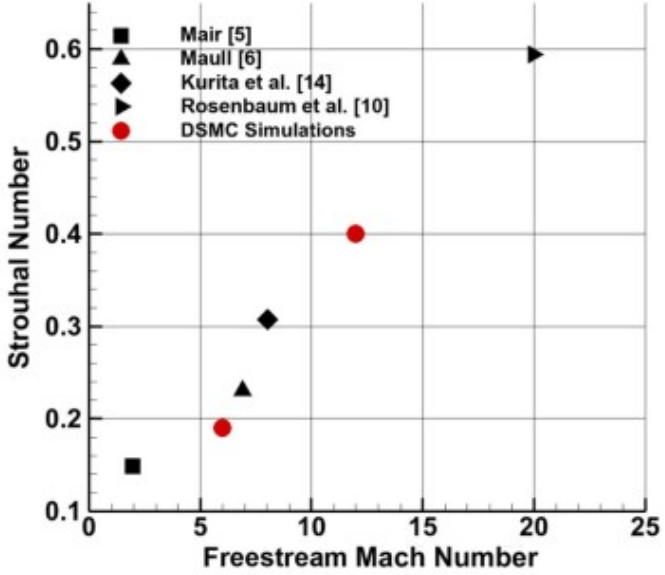


Fig. 10 Comparison of Strouhal number *St* obtained from this work (red circles) with the literature. Note that unit freestream Reynolds numbers *Re* (1/m) are on the order of 10^7 for Mair [5], Maull [6], and Kurita et al. [14]; and they are on the order of 10^5 for Rosenbaum et al. [10] and this work.

5. Discussion

Important points from that study:

1) For cold side-jet flows, DSMC results more or less agree with experiments
2) In downstream, jet and freestream flows do not mix; as a result heat transfer happens from surface to flow. In other words, surface “cools off”.
3) Total drag decreases; normal force increases with the effect of side-jet.
4) If the incoming boundary layer is laminar, frequency of unsteadiness increases with Mach number but does not strongly depend to Reynolds number.

What is the limitations of the study?

1) Only one type of geometry (axisymmetric cone)
2) Only two Ma numbers, 6 and 12
3) Only cold jet is studied, there is no hot-jet case
4) No chemical reactions, no thermal non-equilibrium (too cold for vibrational excitation)
5) Only one jet position condition
6) Only one jet strength condition

6. Possible Future Work

How this study can be improved:

1) Same study with CFD to show accuracy/cost scale,
2) Thermal non-equilibrium/chemical equilibrium N_2 with same setup,
3) Thermal and chemical non-equilibrium N_2 with same setup,
4) Cold-jet with air, thermal and chemical equilibrium,
5) Cold-jet with air, thermal and chemical non-equilibrium,
6) Different jet locations with cold-jet,
7) Different cone half-angles with same cold-jet and setup,
8) Ma 6-7-8-9-10-11-12 to develop a curve fit for amplification factors

What is not worth investigating:

1) Different geometries like flat plate, cone, wedge [1],
2) Flat plate unsteadiness effects [2],
3) Different jet-strengths [3]

[1] Min, B.-Y., Lee, J.-W., & Byun, Y.-H. (2006). Numerical investigation of the shock interaction effect on the lateral jet controlled missile. Aerospace Science and Technology, 10, 385–393.

[2] Miller, W. A., Medwell, P. R., Doolan, C. J., & Kim, M. (2018). Numerical investigation of a pulsed reaction control jet in hypersonic crossflow. Physics of Fluids, 30, 106108. https://doi.org/10.1063/1.5049391

[3] Karpuzcu, I. T., Levin, D. A., Mamrol, D. V., Wagner, L. N., Noftz, M. E., Jewell, J. S., and Smith, S. T., "Jet Flow – Shockwave Interactions in a Hypersonic Flow Using Experimental and Kinetic Methods," AIAA SciTech Forum, San Diego, CA & Virtual, Jan. 3–7, 2022.

Thanks for Listening